MULTI-LABEL TEXT CATEGORIZATION USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

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ABSTRACT

Multi-label text categorization is a Natural Language Processing (NLP) and Machine Learning approach used to assign multiple categories to a single text document. Unlike traditional classification systems that assign only one label, this system can predict more than one category simultaneously.

In this project, TF-IDF vectorization and Logistic Regression are used to process and classify text data into categories such as Technology, Education, Health, and Sports.

INTRODUCTION

Due to the rapid increase in digital content, organizing and classifying text data has become an important task. Many documents belong to more than one category, making multi-label classification necessary.

This project develops a system that processes text data, converts it into numerical format, and predicts multiple relevant categories using machine learning techniques.

OBJECTIVES

- To build a multi-label text classification system
- To apply NLP techniques for text processing
- To convert text into machine-readable features
- To predict multiple categories for a single document
- To improve automated content organization

TECHNOLOGIES USED

Python, Pandas, NumPy, Matplotlib, Scikit-learn, NLP

METHODOLOGY

The project follows these steps:

- Data collection of text documents with labels
- Data preprocessing and cleaning
- Label transformation using MultiLabelBinarizer
- Feature extraction using TF-IDF vectorization
- Splitting dataset into training and testing sets
- Model training using One-vs-Rest Logistic Regression
- Model evaluation using accuracy and classification report
- Prediction of categories for new text

DATASET DESCRIPTION

The dataset contains text documents with multiple labels. Common categories include Technology, Education, Sports, and Health. Each document may belong to one or more categories, making it a multi-label classification problem.

IMPLEMENTATION

The system is implemented using Python and Scikit-learn. TF-IDF is used to convert text into numerical features, and Logistic Regression is applied using a One-vs-Rest strategy for multi-label classification. The model is trained on labeled data and tested on unseen text samples.

RESULTS

The system successfully classifies text into multiple categories. It predicts relevant labels for unseen data and provides classification performance reports. It also helps in understanding keyword importance and category distribution.

ADVANTAGES

- Automates text classification process
- Supports multiple labels per document
- Reduces manual effort
- Works efficiently on large datasets

APPLICATIONS

- News categorization

- Email filtering
- Social media analysis
- Document management systems
- Recommendation systems

FUTURE ENHANCEMENTS

- Deep learning models like BERT and LSTM
- Real-time classification system
- Web-based deployment
- Support for multilingual text

CONCLUSION

The project demonstrates the effectiveness of Machine Learning and NLP in multi-label text classification. Using TF-IDF and Logistic Regression, the system successfully predicts multiple categories for a single document and improves automated text organization.

REFERENCES

- Scikit-learn Documentation
- Python Documentation
- Pandas Documentation
- Research papers on multi-label classification and NLP

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